DUSHMANTHA HERATH

SOFTWARE ENGINEERING UNDERGRADUATE

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github.com/DushmanthaHerath1 dushmantha.vercel.app

PROFESSIONAL SUMMARY

Software Engineering undergraduate specialising in AI engineering, with a full-stack foundation in React, FastAPI and PostgreSQL. Built a LangGraph-based support and refund agent with human-in-the-loop safeguards and deterministic guardrails to reduce LLM unreliability, and a payment-integrated e-commerce platform. Currently conducting undergraduate research on hybrid LLM-deterministic architectures for reliable automated quote generation. Seeking an AI engineering internship to build dependable, production-minded LLM applications.

EDUCATION

Open University of Sri Lanka

2024 - Present

Bachelor of Software Engineering (Hons)

software engineering fundamentals, object-oriented design, data structures and algorithms, database systems, design patterns, systems analysis and design, full-stack development.

University of Kelaniya

2022 — 2025

Bachelor of Arts

Second Class Upper Division | GPA: 3.31

Core coursework focused on programming fundamentals, object-oriented programming, database management systems, web development, mobile application development, computer networks, systems analysis.

University of Kelaniya

2022 — 2025

Diploma in Information Technology

Programming and software development, databases, web development, advanced programming, UX/UI design and information systems; strong performance in programming and database modules.

PROJECTS

AI Support & Refund Agent with Human-in-the-Loop

Python, FastAPI, LangGraph, Groq, PostgreSQL, SQLAlchemy, React 19, Vite, Tailwind CSS, SSE, Pydantic

https://github.com/DushmanthaHerath1/support-ticket-agent

Designed a deterministic customer support pipeline utilizing Groq for high-throughput ReAct tool-calling and Gemini for customer-facing responses, structured ticket classification, and resolution audits with SQLAlchemy and Pydantic v2. Mitigated hallucinations by preloading order context prior to model execution, and integrated LangGraph PostgreSQL checkpoints with interrupt triggers to mandate supervisor review on refund operations. Token streams and live tool-call statuses are delivered asynchronously via FastAPI Server-Sent Events to a React 19 chat and supervisor interface.

Crumb & Crust - Bakery E-Commerce & Operations Management

React 19, FastAPI, Python, SQLAlchemy, PostgreSQL, Alembic, Zustand, Tailwind CSS, PayPal API, JWT

https://github.com/DushmanthaHerath1/Crumb-and-Crust

Architected a full-stack ordering platform using FastAPI and React 19 with Zustand state management, centered on an automated scheduling engine that enforces lead times, blackout dates, checkout-hold expirations, and daily capacity limits to prevent overbooking. Integrated PayPal Orders v2 with server-side capture verification and UUID idempotency keys to eliminate race conditions and duplicate transactions, supported by an OAuth2/JWT-secured kitchen portal for date-based order filtering, real-time inventory toggling, and status tracking.

CV Engine - CV Crafting Application

React 19, Tailwind CSS, React PDF, Zod, @dnd-kit

https://github.com/DushmanthaHerath1/CV-Creator.git

Developed a privacy-first, client-side web application generating ATS-optimized resumes using an extensible five-template registry architecture. Deployed the application commercially within a CV-writing business, driving over LKR 50,000 in revenue through iterative, feedback-driven updates, with active plans to migrate core logic to a FastAPI backend for server-side processing and AI-assisted content generation.

MediConnect - Healthcare Management System (University Group Project)

React 19, TypeScript, Tailwind CSS, TanStack Query, Node.js, Express, PostgreSQL (Prisma)

https://github.com/MediConnectGroupProject/MediConnect.git

Group leader and lead frontend developer; delivered appointment scheduling, patient records and inventory tracking modules.

UNDERGRADUATE RESEARCH

Design and Evaluation of a Hybrid LLM-Deterministic Architecture for Reliable Automated Quote Generation in B2B SMEs (in-progress)

Conducting research grounded in a 22-paper literature review to develop a practical alternative to complex enterprise Configure-Price-Quote (CPQ) systems for resource-constrained SMEs. The proposed architecture delegates natural language comprehension to LLMs while isolating business logic, such as pricing rules, tax computations, customer-specific discounts, and quantity thresholds within deterministic execution engines, reserving human oversight strictly for edge cases.

SKILLS

Python for AI · LLM fundamentals · prompt engineering · LangChain · LangGraph · Google Gemini API · FastAPI · REST APIs · JWT/OAuth2 · Pydantic · Server-Sent Events · React.js · JavaScript · Tailwind CSS · Zustand · PostgreSQL · SQLAlchemy · Java

TOOLS

Git · GitHub · VS Code · IntelliJ IDEA · Figma

CERTIFICATES

Python Intermediate 2024

SoloLearn

www.sololearn.com/certificates/CC-TRFA3WIE

Java Intermediate 2024

SoloLearn

www.sololearn.com/certificates/CC-Z4YNPPYF

LANGUAGES

English (Fluent) Sinhala (Native)

REFERENCES

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